



PRODUCT NAME

High-performance Hand Sanitizer Gel Formula

Description

alcohol-based hand sanitizer gel

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	24 months
Batch Size:	100-500 kg
Processing Time:	2-4 hours
Temperature:	Room temperature
Storage Conditions:	Cool, dry place
Equipment:	Standard mixer
Certification:	Meets industry standards

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Ethanol	Alcohol	71.43%	Active ingredient for microbial control
2	Purified Water	Aqua	22.45%	Solvent
3	Carbomer	Carbomer	0.51%	Thickening agent
4	Triethanolamine	Triethanolamine	0.31%	pH adjuster
5	Glycerin	Glycerin	1.53%	Humectant
6	Isopropyl Myristate	Isopropyl Myristate	1.02%	Emollient
7	Propanediol	Propanediol	2.04%	Humectant and skin conditioning agent
8	Fragrance	Fragrance	0.51%	Scent
9	Phenoxyethanol	Phenoxyethanol	0.2%	Preservative
Total			100%	

Manufacturing Process

Phase 1: Manufacturing Phase

Phase 2: Manufacturing Phase

Phase 3: Manufacturing Phase

Phase 4: Manufacturing Phase

Phase 5: Manufacturing Phase

Phase 6: Manufacturing Phase

Phase 7: Manufacturing Phase

Required Equipment

Standard mixer

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Cool, dry place

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Meets industry standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 100-500 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply as needed according to product instructions